

Memory Devices (Part-2)

MD. AZMAIN YAKIN SRIZON
LECTURER, DEPT. OF CSE, RUET

Modules

- A. Module-1: Read-Only Memories
- B. Module-2: ROM Architecture
- C. Module-3: ROM Timing
- D. Module-4: Types of ROMs
- E. Module-5: Flash Memory
- F. Module-6: ROM Applications

Module-1

Read-Only Memories

Read-Only Memories

- The read-only memory is a type of **semiconductor memory** designed to hold data that either are **permanent or will not change frequently**.
- **During normal operation, no new data can be written** into a ROM, but data **can be read** from ROM.
- For some ROMs, the data that are stored must be built-in during the **manufacturing process**; for other ROMs, the data can be **entered electrically**.
- The process of entering data is called **programming or burning-in** the ROM.
- Some ROMs **cannot have their data changed** once they have been programmed; others can be **erased and reprogrammed** as often as desired.

ROM Block Diagram

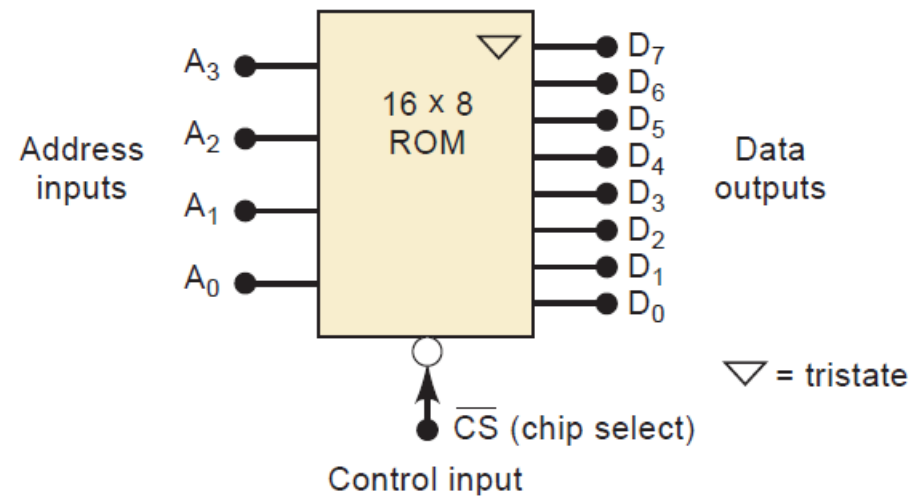


Fig. 1.1: Typical ROM block symbol.

The Read Operation

Word	Address				Data							
	A ₃	A ₂	A ₁	A ₀	D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
0	0	0	0	0	1	1	0	1	1	1	1	0
1	0	0	0	1	0	0	1	1	1	0	1	0
2	0	0	1	0	1	0	0	0	0	1	0	1
3	0	0	1	1	1	0	1	0	1	1	1	1
4	0	1	0	0	0	0	0	1	1	0	0	1
5	0	1	0	1	0	1	1	1	1	0	1	1
6	0	1	1	0	0	0	0	0	0	0	0	0
7	0	1	1	1	1	1	1	0	1	1	0	1
8	1	0	0	0	0	0	1	1	1	1	0	0
9	1	0	0	1	1	1	1	1	1	1	1	1
10	1	0	1	0	1	0	1	1	1	0	0	0
11	1	0	1	1	1	1	0	0	0	1	1	1
12	1	1	0	0	0	0	1	0	0	1	1	1
13	1	1	0	1	0	1	1	0	1	0	1	0
14	1	1	1	0	1	1	0	1	0	0	1	0
15	1	1	1	1	0	1	0	1	1	0	1	1

Word	Address				Data
	A ₃	A ₂	A ₁	A ₀	D ₇ –D ₀
0		0			DE
1		1			3A
2		2			85
3		3			AF
4		4			19
5		5			7B
6		6			00
7		7			ED
8		8			3C
9		9			FF
10		A			B8
11		B			C7
12		C			27
13		D			6A
14		E			D2
15		F			5B

Fig. 1.2: Table showing binary data at each address location (left), the same table in hex (right).

The Read Operation

- In order to read a data word from ROM, we need to do two things:
 - (1) **apply the appropriate address inputs** and then
 - (2) **activate the control** inputs.

Review Questions

- ❖ **True or false:** All ROMs are nonvolatile.
- ❖ Describe the procedure for **reading from ROM**.
- ❖ What is **programming or burning-in** a ROM?

Module-2

ROM Architecture

ROM Architecture

- The internal architecture (structure) of a **ROM IC is very complex**, and we need not be familiar with all of its detail.
- It is instructive, however, to look at a simplified diagram of the internal architecture.
- There are four basic parts: **register array, row decoder, column decoder, and output buffers**.

ROM Architecture

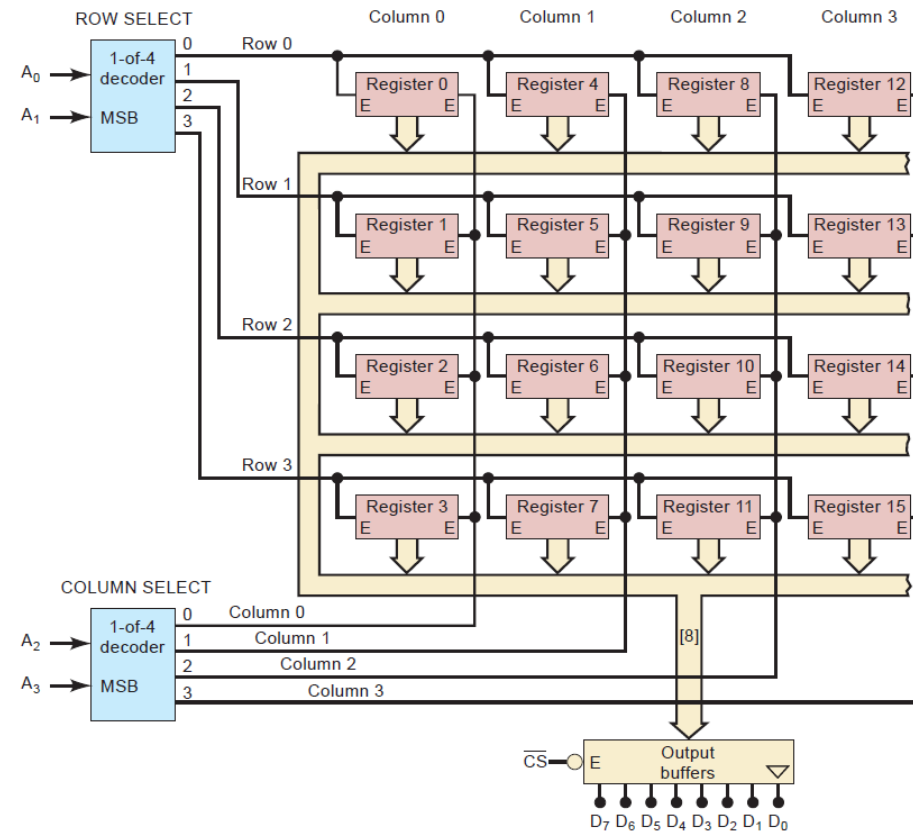


Fig. 2.1: Architecture of a 16×8 ROM. Each register stores one eight-bit word.

Register Array

- The register array **stores the data** that have been programmed into the ROM.
- Each register contains several memory cells equal to the **word size**.
- In this case, each register stores **an eight-bit word**.
- The registers are arranged in a **square matrix array** that is common to many semiconductor memory chips.
- We can specify the position of each register as being in a **specific row and a specific column**.
- For **example**, register 0 is in row 0, column 0, and register 9 is in row 1, column 2.

Address Decoders

- The applied address code $A_3A_2A_1A_0$ determines which register in the array will be enabled to place its eight-bit data word onto the bus.
- Address bits A_1A_0 are fed to a 1-of-4 decoder that activates **one row-select line**, and address bits A_3A_2 are fed to a second 1-of-4 decoder that activates **one column-select line**.

Example 2.1

Question: Which register will be enabled by input address **1101**?

Solution:

$A_3A_2 = \mathbf{11}$ will cause the column decoder to activate the **column 3 select line**,
and $A_1A_0 = \mathbf{01}$ will cause the row decoder to activate the **row 1 select line**.

Example 2.2

Question: What input address will enable **register 7**?

Solution:

The enable inputs of this register are connected to the row 3 and column 1 select lines, respectively.

To select **row 3**, the A_1A_0 inputs must be at **11**, and to select **column 1**, the A_3A_2 inputs must be at **01**.

Thus, the required address will be $A_3A_2A_1A_0 = \mathbf{0111}$.

Output Buffers

- The register that is enabled by the **address inputs will place its data on the data bus.**
- These data **feed into the output buffers**, which will pass the data to the external data outputs, provided that $\overline{CS}$ is LOW.

Example 2.3

Question: Describe the internal architecture of a ROM that stores **4K bytes** and uses a square register array.

Solution:

4K bytes is 4096 eight-bit words. Each word can be thought of as being stored in an eight-bit register, and there are 4096 registers connected to a common data bus internal to the chip.

Because **$4096 = 64^2$** , the registers are arranged in a array; that is, there are 64 rows and 64 columns.

This requires a **1-of-64 decoder** to decode six address inputs for the **row select**, and a second 1-of-64 decoder to decode six other address inputs for the **column select**.

Thus, a total of **12 address inputs is required**. This makes sense because $2^{12} = 4096$, and there are 4096 different addresses.

Review Questions

- ❖ What input address code is required if we want to read the data from **register 9 in Figure 2.1**?
- ❖ Describe the function of the row-select decoder, the column-select decoder, and the output buffers in the ROM architecture.

Module-3

ROM Timing

ROM Timing

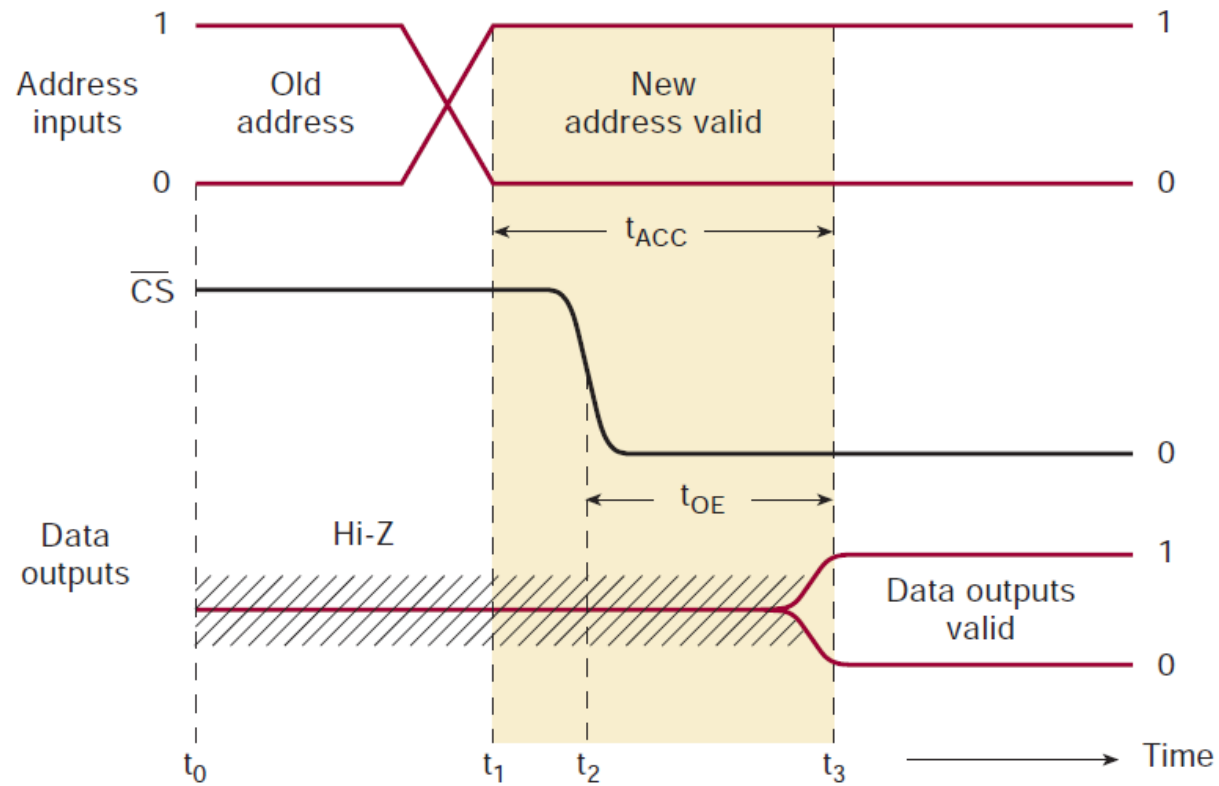


Fig 3.1: Typical timing for a ROM read operation.

Module-4

Types of ROMs

Types of ROMs

- Now that we have a general understanding of the internal architecture and external operation of ROM devices, we will look at the various types of ROMs to see how they differ in the way they are programmed, erased, and reprogrammed.
- Mask-Programmed ROM (**MROM**)
- Programmable ROMs (**PROMs**)
- Erasable Programmable ROM (**EPROM**)
- Electrically Erasable PROM (**EEPROM**)
- **CD-ROM**

Mask-Programmed ROM

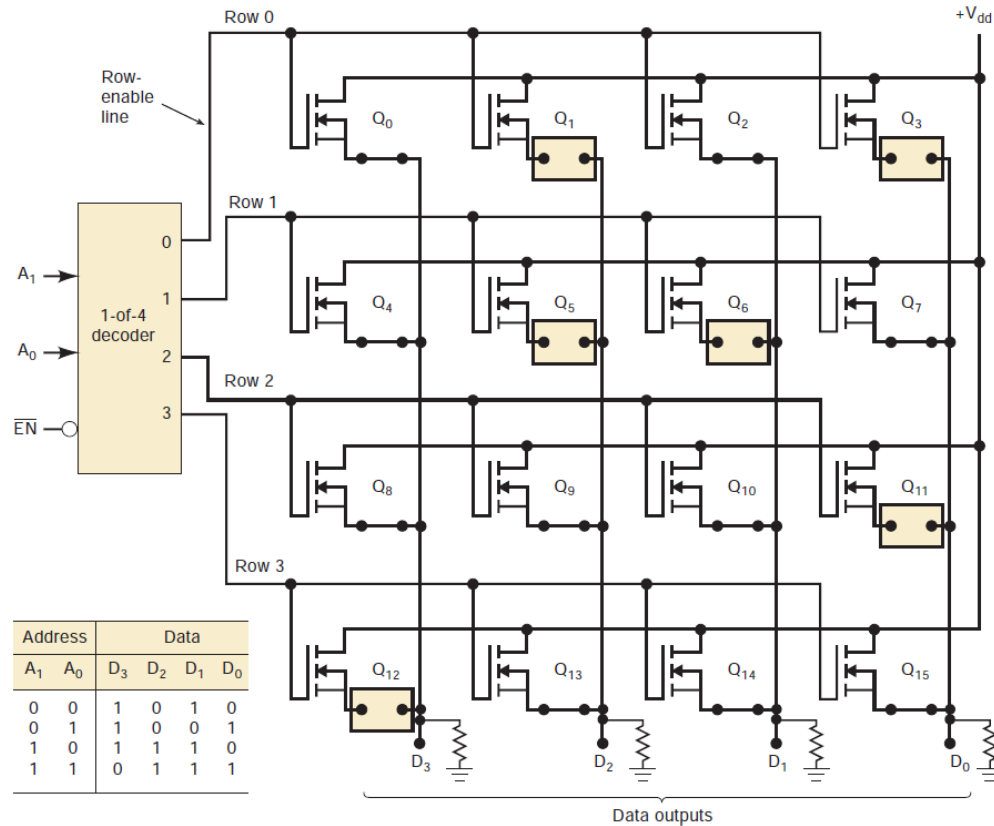


Fig 4.1: Structure of a MOS MROM shows one MOSFET used for each memory cell. An open source connection stores a “0”; a closed source connection stores a “1”.

Programmable ROMs (PROMs)

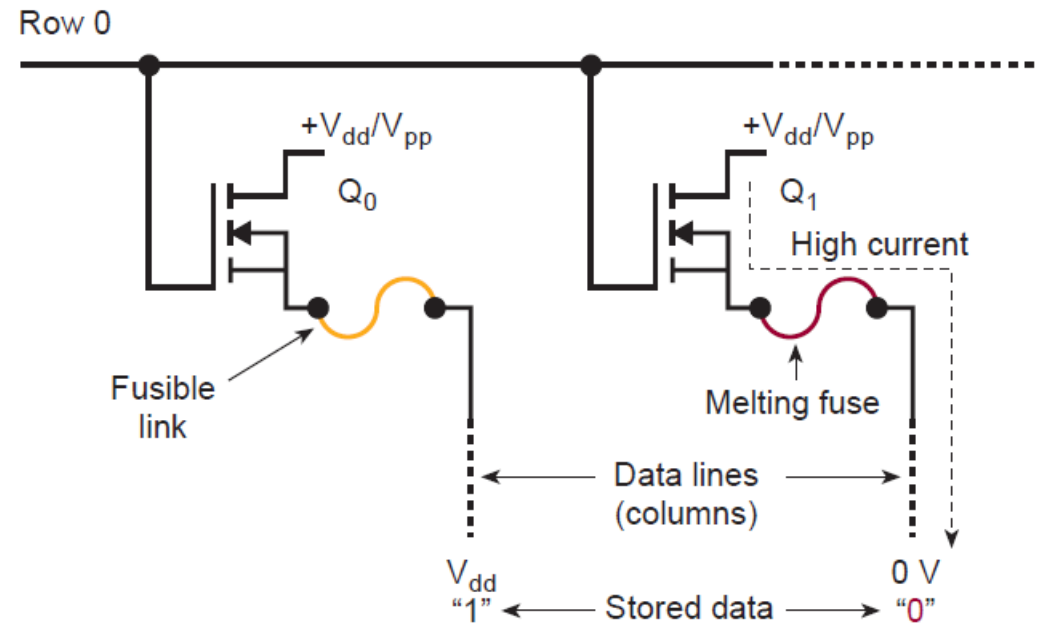
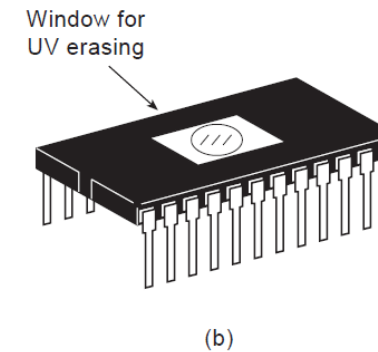
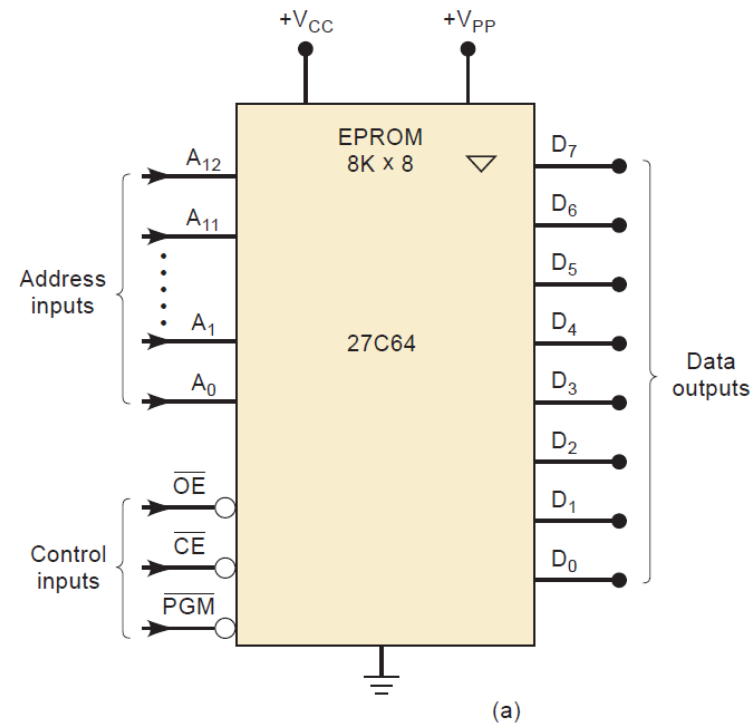


Fig 4.2: PROMs use fusible links that can be selectively blown open by the user to program a logic 0 into a cell.

Erasable Programmable ROM (EPROM)



	Inputs				Outputs
Mode	$\overline{CE}$	$\overline{OE}$	$\overline{PGM}$	V_{PP}	$D_7 - D_0$
Read	0	0	1	0–5V	$DATA_{out}$
Output Disable	0	1	1	0–5V	High Z
Standby	1	X	X	X	High Z
Program	0	1	0	12.75 V	$DATA_{in}$
PGM Verify	0	0	1	12.75 V	$DATA_{out}$

Fig 4.3: (a) Logic symbol for 27C64 EPROM; (b) typical EPROM package showing ultraviolet window; (c) 27C64 operating modes.

Electrically Erasable PROM (EEPROM)

- The disadvantages of the EPROM were overcome by the development of the electrically erasable PROM (EEPROM) as an **improvement** over the EPROM.
- The EEPROM retains the same floating-gate structure as the EPROM, but with the addition of a **very thin oxide region** above the drain of the MOSFET memory cell.
- This modification produces the EEPROM's major characteristic— **its electrical erasability**.

CD-ROM

- A very prominent type of read-only storage used today in computer systems is the **compact disk (CD)**.
- The disks are manufactured with a **highly reflective surface**.
- To store data on the disks, a very intense laser beam is **focused on a very small point** on the disk.
- This beam **burns a light-diffracting pit** at that point on the disk surface.
- Digital data (1s and 0s) are stored on the disk one bit at a time by **burning or not burning a pit** into the reflective coating.
- The digital information is arranged on the disk as a continuous **spiral of data points**.
- The precision of the laser beam allows very large quantities of data (**over 550 Mbytes**) to be stored on a small, **120-mm disk**.

Review Questions

- ❖ **True or false:** An MROM can be programmed by the user.
- ❖ How does a PROM differ from an MROM? Can it be erased and reprogrammed?
- ❖ What EPROM **shortcomings are overcome** by EEPROMs?
- ❖ What are the **major drawbacks of EEPROM**?

Module-5

Flash Memory

ROM Applications

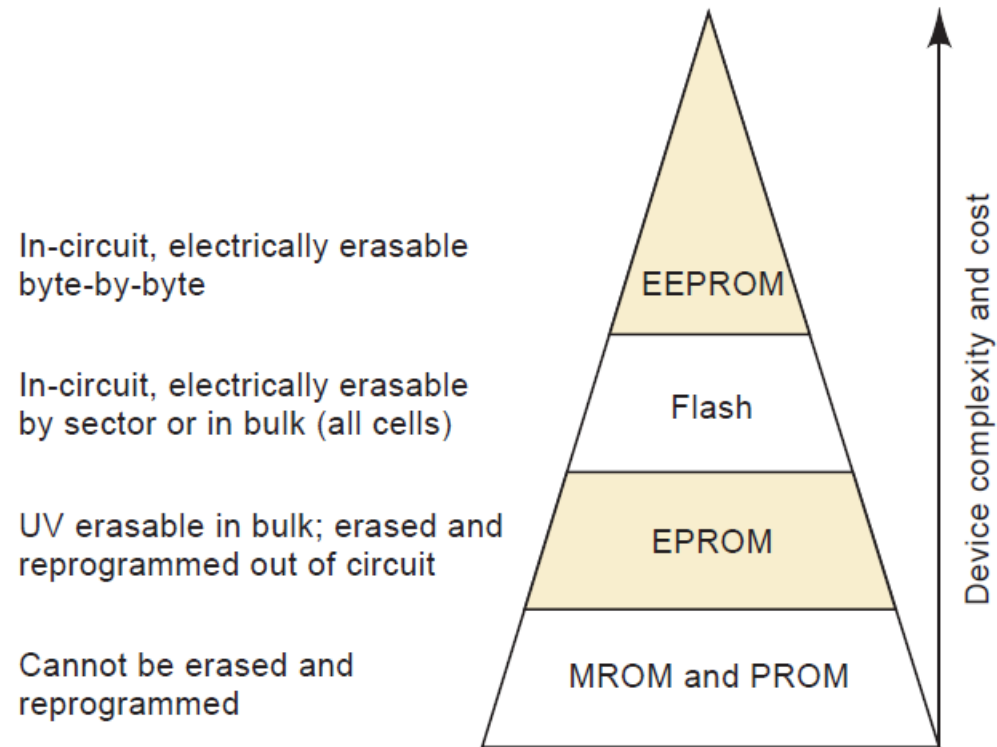


Fig 5.1: Trade-offs for semiconductor nonvolatile memories show that complexity and cost increase as erase and programming flexibility increases.

Module-6

ROM Applications

ROM Applications

- Embedded Microcontroller Program Memory
- Data Transfer and Portability
- Bootstrap Memory
- Data Tables
- Data Converter
- Function Generator
- Auxiliary Storage

References

Digital Systems by Tocci, Widmer, Moss

Chapter 12: Memory Devices (12.4-12.9)

Pages: 795-814

Thank You

Next Topics

Random Access Memory